GRADE 9
PRE-TECHNICAL STUDIES

DETAILED FULL-YEAR LEARNER NOTES

Terms 1, 2 and 3 | Concepts • Worked examples • Practical tasks • Revision

Junior School Learning Hub
Original teacher-prepared learning resource

How to use these notes

Pre-Technical Studies combines practical, technical, digital and entrepreneurial learning. Read the explanations, study the examples, practise drawing and planning, and complete activities under appropriate supervision. Keep a portfolio of sketches, designs, programming work, project records and reflections.

Safety is essential. Do not climb raised platforms, handle unknown chemicals, operate power tools, connect electrical circuits or use workshop machinery without direct supervision by a trained adult. Follow school workshop rules and product safety instructions.

Curriculum alignment note: These notes follow the Grade 9 KICD Pre-Technical Studies design strands and sub-strands: Foundations of Pre-Technical Studies; Communication in Pre-Technical Studies; Materials for Production; Tools and Production; and Entrepreneurship. This is an original teacher-prepared resource, not an official KICD publication. Verify terminology and lesson emphasis against the current curriculum design used by your school.

Learning area overview

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The lesson counts are suggested allocations in the curriculum design. Schools may adjust pacing to suit their context while ensuring the intended learning outcomes are addressed.

STRAND 1.0: FOUNDATIONS OF PRE-TECHNICAL STUDIES

1.1 Safety on Raised Platforms

A raised platform is a working surface above the surrounding ground or floor. Examples include ladders, scaffolds, step platforms, mezzanine floors, loading platforms and elevated work stages. Raised platforms help people reach high places, inspect structures, paint, install fittings or carry out maintenance.

Working at height can cause serious injury if a person falls or if objects fall onto people below. Risk increases when a platform is unstable, overloaded, wet, damaged or used in strong wind. A safe work plan begins by asking whether the task can be done from ground level. If working at height is necessary, the correct access equipment and protection must be selected by a competent adult.

Key points

- Inspect the platform, ladder, steps, guardrails, locking devices and supporting surface before use.
- Place access equipment on firm, level ground and secure it as instructed by the manufacturer.
- Never exceed the stated load capacity; include the worker, tools and materials in the load.
- Keep platforms free of clutter, oil, water and loose materials.
- Use guardrails, toe boards, fall-arrest equipment or other protection where required and properly fitted.
- Keep people away from the area below; secure tools and materials to prevent falling objects.
- Do not use a damaged ladder or platform. Label it and report it for repair or removal.
- Do not climb in unsafe weather or when tired, dizzy or unwell.

Hazard	Possible consequence	Control measure
Unstable ladder or platform	Fall and serious injury	Use suitable inspected equipment on a firm surface; secure it.
Overloading	Collapse or loss of balance	Follow the safe working load and avoid carrying excessive materials.
Wet or cluttered surface	Slip or trip	Keep the platform clean, dry and organised.
Objects falling	Injury to people below	Use tool lanyards where suitable and restrict access below.
Unsafe access	Fall while climbing	Maintain safe access and use three points of contact on a ladder.

Worked example / application

A learner needs to observe a roof gutter. The learner should not climb onto a chair or improvise a platform. The safe action is to inform the teacher and allow a trained adult to select appropriate access equipment and fall-prevention measures.

Practical activity

Use photographs or a supervised walk-around to identify raised platforms in the school. For each one, list its purpose, possible hazards and visible safety features. Do not climb or test equipment.

Check your understanding

1. Define a raised platform and give four examples.

2. State five checks to make before a raised platform is used.

3. Why should people be kept away from the area below elevated work?

4. Explain why a chair or desk is not a safe substitute for a ladder.

Suggested answers

1. A working surface above ground level; examples include ladder, scaffold, mezzanine, step platform and loading platform.

2. Check stability, damage, footing, locking devices, guardrails, load capacity and weather conditions.

3. Tools or materials may fall and injure people below.

4. Furniture is not designed or rated for climbing and may tip, break or slide.

1.2 Handling Hazardous Substances

A hazardous substance is a material that can harm people, animals, property or the environment because of its chemical or physical properties. Examples include corrosive cleaners, fuels, solvents, paints, pesticides, acids, alkalis, compressed gases and some workshop dusts.

Exposure can occur by inhalation (breathing in), skin or eye contact, swallowing, or injection through cuts or punctures. The effect depends on the substance, amount, route and duration of exposure. Some substances cause immediate burns or irritation; others may cause long-term harm.

Key points

- Read the label and safety instructions before a substance is used. Identify hazard symbols and precautionary statements.
- Use substitution: where possible, choose a less hazardous product for the same task.
- Use suitable ventilation and keep containers closed when not in use.
- Wear the protective equipment specified for the task, such as suitable gloves, goggles or apron.
- Never taste, smell directly or mix unknown substances. Some combinations release toxic gases or heat.
- Keep chemicals in original labelled containers. Never use food or drink containers for chemicals.
- Store incompatible substances separately and secure them away from learners, food and animal feed.
- Clean spills only when trained and equipped; report spills immediately to the responsible adult.
- Dispose of chemical waste according to the label and local rules; never pour unknown chemicals into drains or onto soil.

Hazard class / symbol idea	Meaning	Precaution
Corrosive	Can burn skin/eyes or corrode materials.	Avoid contact; use specified protection and adult supervision.
Flammable	Can catch fire easily.	Keep away from flames, sparks

		and heat; store correctly.
Toxic / poisonous	Can cause serious harm if inhaled, swallowed or absorbed.	Avoid exposure; secure storage and follow label instructions.
Irritant / harmful	May irritate skin, eyes or breathing passages.	Avoid contact and use suitable ventilation/protection.
Gas under pressure	Container may burst if heated; gas may cause other hazards.	Protect cylinder from heat and physical damage; secure upright as specified.

Worked example / application

If a cleaning liquid splashes onto a learner's skin or eyes, the learner should alert the teacher immediately and follow the product's first-aid instructions. Do not try to neutralise a chemical on the skin with another chemical. The responsible adult should consult the label or safety data sheet and seek medical help where necessary.

Practical activity

Study safe, empty product labels or printed examples. Identify the product name, hazard warnings, precautionary statements, first-aid advice and storage instructions. Do not open or handle hazardous products during the activity.

Check your understanding

1. What is a hazardous substance?

2. Name four routes by which a substance can enter the body.

3. Why should chemicals remain in their original containers?

4. State five safe storage practices.

5. Why is mixing unknown cleaning products dangerous?

Suggested answers

1. A material whose properties can harm people, animals, property or the environment.

2. Inhalation, skin/eye contact, swallowing and injection through cuts/punctures.

3. The original label identifies the contents, hazards, precautions and first-aid instructions.

4. Keep labelled, closed, secure, separated from incompatible products, away from food and out of children's reach.

5. A reaction may release toxic gases, heat or corrosive products.

1.3 Self-Exploration and Career Development

Self-exploration is the process of understanding one's interests, talents, abilities, values, strengths and areas for improvement. It helps a learner make informed decisions about learning and future work.

A talent is a natural or developed aptitude for an activity. An ability is the capacity to perform a task, often improved through practice. A skill is an ability developed through learning and experience. Interests are activities or subjects that a person enjoys. Values are principles that guide choices, such as honesty, responsibility, creativity and service.

Career development is a continuous process of exploring opportunities, building relevant skills, setting goals and making informed choices. A career pathway may include education, training, work experience and progression into related roles.

Key points

- Identify interests and strengths through reflection, feedback and participation in activities.
- Try different tasks in clubs, projects, sports, creative activities and practical lessons.
- Set realistic short-term and long-term goals and review progress.
- Research entry requirements, training institutions, working conditions and career opportunities.
- Develop transferable skills such as communication, teamwork, problem-solving, digital literacy and time management.
- Use talents ethically: respect others, avoid cheating, plagiarism, discrimination and misuse of digital tools.

Interest / ability	Possible career pathways (examples)
Drawing and visualisation	Technical drawing, architecture, product design, illustration.
Practical making and repair	Carpentry, electrical installation, plumbing, mechanical work.
Computing and logical thinking	Software development, ICT support, digital design, data work.
Organising and communicating	Entrepreneurship, project coordination, sales, administration.
Helping and caring	Teaching, health support, community development.

Worked example / application

A learner who enjoys drawing, measuring and making models may explore technical design or construction-related careers. The learner should research the required subjects, training, safety responsibilities and professional standards rather than choosing a career based only on its name.

Practical activity

Prepare a personal skills map with three interests, three abilities, two values, one area to improve and two career pathways to research. Share only information you are comfortable discussing.

Check your understanding

1. Differentiate a talent from a skill.

2. List five factors to consider when exploring a career.

3. Give four transferable skills useful in technical careers.

4. Why should career decisions be based on research and self-awareness?

Suggested answers

1. A talent is an aptitude that may come naturally or develop; a skill is learned and improved through practice.

2. Interests, abilities, values, training requirements, opportunities, working conditions and personal goals.

3. Communication, teamwork, problem-solving, digital literacy and time management.

4. It helps match personal strengths and values with realistic training and work requirements.

STRAND 2.0: COMMUNICATION IN PRE-TECHNICAL STUDIES

2.1 Oblique Projection

Technical drawing communicates the shape, size and features of an object using agreed drawing methods. Oblique projection is a pictorial drawing method in which the front face of an object is shown in its true shape and size while depth lines are drawn at an angle, commonly 45 degrees.

An oblique drawing helps a viewer understand a three-dimensional object on a two-dimensional page. It is useful for communicating the appearance of simple objects such as boxes, blocks, furniture components and machine parts.

Key points

- Choose the front face that shows the most important shape or detail.
- Draw the front face in its true shape and size using light construction lines.
- Draw parallel receding lines from the appropriate corners at the chosen angle, often 45 degrees.
- Mark the depth consistently along the receding lines.
- Join the ends to complete the visible faces; use darker lines for visible outlines.
- Use hidden detail only when required by the drawing task; avoid unnecessary lines.
- Add dimensions and labels clearly where requested.

Drawing type	Front face	Depth representation
Oblique projection	True shape and size	Receding lines drawn at an angle, often 45°
Isometric drawing	Axes are equally inclined in the conventional isometric arrangement	Three principal directions represented consistently
Orthographic drawing	Separate 2D views such as front, plan and side	Depth is communicated through multiple views

Worked example / application

To draw a rectangular block 60 mm wide, 40 mm high and 30 mm deep: draw a 60 mm by 40 mm front rectangle. From its depth corners draw parallel 45° receding lines representing 30 mm depth using the chosen oblique convention. Join the corresponding endpoints to complete the top and side faces. Use a ruler, set square or protractor as available.

Practical activity

Draw a simple rectangular box, book or desk organiser in oblique projection. Label the front face, top face, side face, width, height and depth. Compare drawings with a partner for parallel lines and consistent depth.

Check your understanding

1. What is oblique projection?

2. Which face is drawn in true shape and size?

3. State four steps in constructing an oblique drawing.

4. A block is 80 mm wide, 50 mm high and 25 mm deep. Which dimensions describe the front face and which describes depth?

Suggested answers

1. A pictorial drawing method showing the front face in true shape and size and depth along angled receding lines.

2. The front face.

3. Draw front face, draw parallel receding lines, mark depth, join endpoints and darken visible outlines.

4. Width 80 mm and height 50 mm describe the front face; 25 mm is depth.

2.2 Visual Programming

Programming is the process of creating instructions that a computer or digital device follows to perform a task. Visual programming uses graphical blocks, icons, flow diagrams or connected symbols instead of relying mainly on typed text code. Examples of beginner-friendly visual environments include block-based coding tools.

An algorithm is a finite, ordered set of steps for solving a problem. A program is an implementation of instructions in a programming environment. A sequence is the order in which instructions run. Selection allows a program to choose between alternatives. Iteration repeats a set of instructions.

Key points

- Identify the problem and describe the expected output.
- Break the problem into smaller steps and plan an algorithm.
- Use sequence blocks to carry out instructions in order.
- Use selection (IF/THEN/ELSE) when an action depends on a condition.
- Use loops to repeat instructions a set number of times or while a condition remains true.
- Use variables to store values that may change, such as score, name or count.
- Test the program with different inputs, find errors (debugging) and improve it.
- Use digital tools responsibly: respect privacy, copyright, passwords and other users.

Programming concept	Meaning	Example
Sequence	Instructions run in order.	Ask name → greet user → display message.
Selection	Choose an action based on a condition.	IF score ≥ 50 THEN show pass ELSE show retry.
Iteration / loop	Repeat instructions.	Repeat a movement 10 times.
Variable	Named storage for a value.	score = 0; increase score after a correct answer.
Event	An action that triggers code.	When start button is clicked, begin the activity.

Debugging	Finding and correcting errors.	Correct a wrong comparison or missing block.

Worked example / application

A simple quiz algorithm: (1) set score to zero; (2) ask a question; (3) if answer is correct, add one to score; otherwise keep score unchanged; (4) repeat for each question; (5) display the final score. The IF condition is selection, while repeating the questions is iteration.

Practical activity

In a block-based environment or on paper, design a short interactive story or quiz. Include a start event, at least one sequence, one selection and one loop. Test the design using two different inputs.

Check your understanding

1. Define an algorithm.

2. Differentiate sequence and iteration.

3. What is a variable?

4. Explain selection using an IF/ELSE example.

5. What is debugging and why is testing important?

Suggested answers

1. A finite ordered set of steps for solving a problem.

2. Sequence runs instructions in order; iteration repeats instructions.

3. A named storage location for a value that can change.

4. IF the answer is correct THEN add a point ELSE display a retry message.

5. Debugging is finding and correcting errors; testing checks whether the program behaves as intended for different inputs.

STRAND 3.0: MATERIALS FOR PRODUCTION

3.1 Wood

Wood is a natural material obtained from the stems and branches of trees. It is widely used in construction, furniture, tools, packaging and craft products. Wood is renewable when forests are managed responsibly and trees are replanted, but uncontrolled harvesting can cause deforestation and loss of habitats.

Timber is wood that has been processed into a form suitable for construction or manufacture. Hardwood and softwood are broad botanical groupings, not simple measures of how hard an individual piece feels. Some hardwoods are relatively soft and some softwoods are strong.

Key points

- Hardwood comes from broad-leaved flowering trees; examples include eucalyptus and mvule. Uses depend on the species and properties.

- Softwood comes mainly from coniferous trees; examples include pine and cypress. It is often used in construction, joinery and furniture.
- Wood properties include grain, colour, density, strength, durability, workability and moisture content.
- Seasoning is the controlled drying of timber to reduce moisture before use. Poorly dried timber may shrink, warp or crack.
- Common defects include knots, shakes, splits, warping, decay and insect attack.
- Wood can be worked by measuring, marking, cutting, planing, drilling, joining, sanding and finishing, using suitable tools and safety procedures.

Property / feature	Meaning and effect
Grain	Direction and arrangement of wood fibres; affects appearance and working.
Density	Mass per unit volume; often related to weight and some strength properties.
Moisture content	Amount of water in wood; changes can cause shrinkage or swelling.
Durability	Ability to resist decay, insects and wear.
Workability	Ease with which wood can be cut, shaped, joined and finished.
Knot	A place where a branch was attached; may affect appearance and strength.

Worked example / application

A shelf intended to carry books should be made from timber of suitable strength, thickness and condition. The designer should consider span, load, support, moisture, joint type and finish. The shelf must be tested safely before use.

Practical activity

Observe labelled wood samples or photographs. Record colour, grain, visible defects and possible uses. Do not cut or sand samples without teacher approval and appropriate protection.

Check your understanding

1. Differentiate wood and timber.

2. What is seasoning and why is it important?

3. Name five properties considered when selecting wood.

4. State four common timber defects.

5. Why should wood be sourced responsibly?

Suggested answers

1. Wood is the natural material from trees; timber is wood processed for use in construction or manufacture.

2. Controlled drying to reduce moisture; it helps reduce later shrinkage, warping and cracking.

3. Grain, density, strength, durability, workability, moisture content and appearance.

4. Knots, shakes, splits, warping, decay and insect attack.

5. Responsible sourcing helps conserve forests, biodiversity and future supplies.

3.2 Handling Waste Materials

Waste materials are unwanted or discarded substances produced by homes, schools, farms, businesses and industries. Some waste can be reused, repaired, recycled or converted into useful products. Poor waste handling can cause injuries, pollution, blocked drainage, disease and loss of valuable resources.

The waste hierarchy prioritises preventing waste, then reducing, reusing, repairing, recycling and recovering value where appropriate. Disposal is the last option for waste that cannot safely be recovered.

Key points

- Separate waste at source into clearly identified categories where a safe collection system exists.
- Reuse materials where they remain safe and suitable, for example clean containers for non-food storage or scrap wood for approved models.
- Recycle materials through appropriate collection systems; not every material is accepted everywhere.
- Keep sharp, broken, contaminated or chemical waste separate and let trained adults manage it.
- Do not burn plastics, treated wood or unknown waste; harmful smoke may be released.
- Keep waste away from drains, rivers, food preparation areas and animal feeding points.
- Use personal protection and handwashing when handling ordinary waste; learners should not handle hazardous waste.

Waste category	Examples	Possible safe action
Paper and cardboard	Clean boxes, paper offcuts	Reuse or send to paper recycling where available.
Metal	Clean metal offcuts, cans	Collect safely for authorised recycling; avoid sharp edges.
Plastic	Clean packaging and containers	Sort by accepted local recycling type; avoid unsafe reuse.
Wood	Untreated offcuts and sawdust	Reuse in suitable projects or manage as approved; avoid burning treated wood.
Organic	Food and plant waste	Compost suitable materials using hygienic methods.
Hazardous	Solvents, batteries, chemical containers, contaminated materials	Keep separate and refer to trained adults and approved disposal channels.

Worked example / application

A class making a model may use clean cardboard offcuts rather than buying new sheets. This prevents waste and reduces cost. The class should reject materials that are dirty, mouldy, sharp, chemically contaminated or structurally unsuitable.

Practical activity

Conduct a supervised waste audit in a classroom. Record the types and approximate quantities of waste generated, then suggest ways to prevent, reuse or recycle suitable materials. Do not open waste bags or handle hazardous items.

Check your understanding

1. Explain the waste hierarchy.

2. Give three benefits of separating waste.

3. Why should unknown chemical containers not be reused?

4. Suggest two ways a school can reduce waste.

Suggested answers

1. Prevent, reduce, reuse, repair, recycle/recover and dispose only as a last resort.

2. Improves recovery, reduces contamination and makes safe handling easier.

3. Residues may remain and cause poisoning, burns or dangerous reactions; labels may also be missing.

4. Use reusable materials, reduce printing, repair items, sort recyclables and compost suitable organic waste.

STRAND 4.0: TOOLS AND PRODUCTION

4.1 Holding Tools

Holding tools grip, support or secure a workpiece so that it can be measured, cut, drilled, shaped or assembled safely and accurately. A workpiece is the material being worked on.

Examples include bench vices, hand vices, clamps, pliers, tongs and clips. The correct tool depends on the shape, size, material and operation. Holding tools should secure the work without damaging it.

Key points

- A bench vice is fixed to a workbench and holds a workpiece between jaws.
- A G-clamp or C-clamp applies pressure to hold material against a bench or another surface.
- Pliers grip, bend or hold small objects; different designs serve different purposes.
- Tongs hold hot or awkward objects where the tool is designed for that purpose and the user is trained.
- Clips and spring clamps hold light materials temporarily.
- Check jaws, handles, threads and moving parts before use; do not use cracked or loose tools.
- Keep fingers clear of pinch points and avoid excessive force that may damage the workpiece or tool.

Tool	Main function	Example use
Bench vice	Holds work firmly on a bench.	Securing a metal or wood workpiece during filing or sawing.
G/C-clamp	Clamps material to a surface or another piece.	Holding two boards during gluing.
Pliers	Grip, bend or hold small items.	Holding a small component

		during assembly.
Hand vice	Holds small workpieces.	Holding a small item for careful work.
Spring clip	Temporary light holding.	Holding paper, thin card or light components.

Worked example / application

When gluing two pieces of wood, apply suitable adhesive, align the pieces and use clamps with protective pads if necessary. Tighten only enough to hold the joint; excessive pressure may distort or damage the wood.

Practical activity

Identify holding tools from actual tools or pictures. For each, name its parts, purpose, suitable workpiece and one safety precaution. Demonstrations must be done by the teacher.

Check your understanding

1. What is the purpose of a holding tool?

2. Differentiate a bench vice and a clamp.

3. State four checks before using a holding tool.

4. Why should excessive clamping pressure be avoided?

Suggested answers

1. To grip or secure a workpiece during an operation.

2. A bench vice is fixed to a bench; a clamp is usually portable and applies pressure to hold items together or to a surface.

3. Check jaws, handles, threads, moving parts and mounting.

4. It can damage the workpiece, distort material or damage the tool.

4.2 Driving Tools

Driving tools transfer force to fasteners or materials. Examples include hammers, mallets, screwdrivers, spanners and punches. Each tool is designed for a particular task; using the wrong tool can damage the fastener, workpiece or user.

A hammer has a head and handle and is used for striking. A mallet usually has a softer or non-metallic head and is used where a metal hammer could damage the surface. A screwdriver turns screws. A spanner turns nuts and bolts.

Key points

- Choose a hammer of suitable type and weight for the task; check that the head is secure.
- Use a screwdriver whose tip matches the screw head and whose size is suitable.
- Use a spanner that fits the nut or bolt correctly; a poor fit can round the edges.
- Keep the workpiece stable using a suitable holding tool.
- Strike away from the body and keep other people outside the working area.
- Do not use a screwdriver as a chisel or pry bar unless it is specifically designed for that purpose.

- Store sharp or pointed driving tools safely and keep handles in good condition.

Tool	Use	Important precaution
Claw hammer	Driving and removing suitable nails.	Keep head secure; use correct nail and controlled strikes.
Mallet	Striking surfaces where a softer impact is needed.	Use the correct mallet material for the workpiece.
Screwdriver	Tightening or loosening screws.	Match tip to screw; do not use as a lever.
Spanner	Turning nuts and bolts.	Use correct size and pull in a controlled direction.
Punch	Marking or driving pins and similar components.	Use correct punch and eye protection where required.

Worked example / application

If a screw has a cross-shaped head, use a suitable cross-head screwdriver that fits fully. A poorly fitting flat screwdriver may slip, damage the screw and injure the hand.

Practical activity

Match pictures of driving tools to their names and uses. Create a tool-care poster showing inspection, cleaning, safe storage and correct selection.

Check your understanding

1. What is a driving tool?

2. Give two differences between a hammer and a mallet.

3. Why must a screwdriver fit the screw head?

4. What can happen if the wrong spanner is used?

Suggested answers

1. A tool used to apply force to a fastener or material.

2. A hammer commonly has a metal head; a mallet has a softer or non-metallic head for less damaging impact.

3. A correct fit reduces slipping and damage to the screw and user.

4. It may slip, round the nut or bolt, damage the tool or cause injury.

4.3 Project: Design, Make, Test and Improve

A project is a planned piece of work carried out to solve a problem or meet a need. In Pre-Technical Studies, a project may involve designing and making a simple product, model or prototype using appropriate materials and tools.

A prototype is an early model used to test a design idea. It may be full-size or reduced-size and may use inexpensive materials. Testing helps identify weaknesses before a final product is made.

Key points

- Identify a real need or problem and describe who the user is.
- Research existing solutions and gather information about materials, dimensions, cost and safety.
- Generate several ideas using labelled sketches; compare strengths and weaknesses.
- Select a design using criteria such as function, safety, cost, durability, appearance and sustainability.
- Prepare a working drawing, materials list, tool list, cost estimate and sequence of operations.
- Make the prototype using safe methods and suitable supervision.
- Test the prototype against clear criteria and record evidence.
- Evaluate results, identify improvements and communicate the final outcome.

Project stage	Questions to answer	Evidence to keep
Identify	What problem am I solving? Who is the user?	Problem statement and user needs.
Research	What solutions already exist? What materials are suitable?	Notes, observations and references.
Design	What are my possible ideas? Which is best and why?	Sketches, dimensions and selected design.
Plan	What tools, materials, time and cost are needed?	Materials list, tool list and work plan.
Make	How will I produce it safely?	Progress notes and photographs/drawings.
Test	Does it meet the criteria?	Test table and observations.
Evaluate	What worked and what should change?	Evaluation and improved design.

Worked example / application

Project idea: a desk organiser made from clean, reusable cardboard. Need: learners require a place to keep pencils and rulers. Criteria: stable base, compartments of suitable size, safe edges, low cost and neat finish. Test by placing stationery inside and checking whether the organiser tips or tears. Improve the base or joints if necessary.

Practical activity

Plan a small school-use product or model with teacher approval. Complete a design brief, at least two sketches, materials and tool lists, safety plan, cost estimate, prototype, test record and evaluation. Use only approved materials and supervised tools.

Check your understanding

1. What is a prototype?

2. List the main stages of a design-and-make project.

3. Why should a design be tested against criteria?

4. Name six items that could be included in a project portfolio.

Suggested answers

1. An early model used to test and improve a design idea.

2. Identify need, research, generate ideas, select design, plan, make, test, evaluate and communicate.

3. Testing provides evidence that the product meets its purpose and reveals areas for improvement.

4. Brief, research notes, sketches, drawings, materials list, tool list, safety plan, cost estimate, test results and evaluation.

STRAND 5.0: ENTREPRENEURSHIP

5.1 Financial Services

Financial services are services that help people and organisations save, borrow, transfer, insure and manage money. They are provided by regulated institutions and service providers such as banks, SACCOs, microfinance institutions, mobile money providers and insurance companies.

Savings are money set aside for future use. A loan is money borrowed that must be repaid under agreed terms. Interest is the cost of borrowing or the return earned on some savings. Insurance is a contract that provides financial protection against specified risks in return for a premium.

Key points

- Deposit accounts help customers keep money and may earn interest depending on the account.
- Payment and transfer services allow money to move between people and businesses.
- Credit facilities provide loans or overdrafts subject to eligibility and repayment conditions.
- SACCOs are member-based cooperative financial organisations; members may save and access credit under the SACCO's rules.
- Insurance may cover risks such as illness, accidents, property loss or business interruption, depending on the policy.
- Compare fees, interest, repayment terms, access, security and provider regulation before choosing a service.
- Protect PINs, passwords and one-time codes. Never share them with callers or strangers.

Service provider	Examples of services	Considerations
Commercial bank	Savings, payments, transfers, loans.	Fees, account terms, access, interest and security.
SACCO	Member savings, loans and other member services.	Membership rules, contributions, loan terms and governance.
Microfinance institution	Savings or credit services depending on licence and product.	Total borrowing cost, repayment schedule and suitability.
Mobile money provider	Wallet, transfers, merchant payments and bill payments.	Transaction charges, limits, fraud protection and PIN safety.
Insurance company	Risk protection under a policy.	Covered events, exclusions, premium and claim requirements.

Worked example / application

A small enterprise needs KSh 5,000 for materials. Before borrowing, the owner should compare the total amount repayable, fees, due dates, penalties and expected cash flow. Borrowing is not automatically profitable; the business must be able to repay without harming essential needs.

Practical activity

Prepare a comparison chart using publicly available information from two financial service products. Do not enter personal account details or conduct real transactions for the activity.

Check your understanding

1. Define financial services.
2. Differentiate savings and a loan.
3. What is insurance?
4. State five factors to consider before taking a loan.
5. Why should a person protect a mobile money PIN and OTP?

Suggested answers

1. Services that help people save, borrow, transfer, insure and manage money.
2. Savings is money set aside; a loan is borrowed money that must be repaid under agreed terms.
3. A contract that provides protection against specified financial risks in exchange for a premium.
4. Total cost, interest, fees, repayment schedule, penalties, affordability and provider legitimacy.
5. They can be used to access an account or authorise transactions; sharing them can lead to theft.

5.2 Government and Business

Government influences business through laws, policies, public services, infrastructure, taxation, regulation and support programmes. These actions can create an environment in which businesses operate safely and fairly.

Businesses produce goods or provide services to meet customer needs. They may be sole proprietorships, partnerships, companies, cooperatives or other lawful forms. Each form has different ownership, liability, registration and management requirements.

Key points

- Government provides infrastructure such as roads, electricity, water systems and communication networks.
- Government establishes laws and standards to protect consumers, workers, the environment and fair competition.
- Taxes and licences help fund public services and regulate certain activities; businesses should comply with applicable requirements.
- Public procurement can create opportunities for eligible businesses under relevant rules.
- Training, extension, enterprise funds and business development services may support entrepreneurs where programmes are available.
- Businesses create employment, pay taxes, supply goods and services, innovate and contribute to community development.

- Ethical businesses avoid bribery, fraud, discrimination, unsafe products, false advertising and environmental harm.

Government role	How it affects business
Infrastructure provision	Reduces transport and operating difficulties.
Regulation and standards	Promotes safety, quality and fair business conduct.
Taxation	Raises public revenue; businesses must understand applicable obligations.
Education and training	Builds skills needed by workers and entrepreneurs.
Consumer protection	Helps protect buyers from unsafe or misleading products.
Enterprise support	May provide training, information, finance programmes or market opportunities.

Worked example / application

A food business may need to follow public health requirements, use hygienic premises, provide safe products and meet applicable licensing and tax obligations. Compliance protects customers and helps build trust.

Practical activity

Choose a small local business and identify three ways government services or rules affect it. Use public information or a teacher-provided case study; do not collect confidential business records.

Check your understanding

1. State five ways government supports or regulates business.
2. Give four contributions businesses make to society.
3. Why are standards and consumer protection important?
4. Give three examples of unethical business practices.

Suggested answers

1. Infrastructure, laws, licensing, taxation, training, consumer protection and enterprise support.
2. Employment, goods/services, tax revenue, innovation and community development.
3. They help ensure safety, quality, accurate information and fair treatment.
4. Bribery, fraud, false advertising, unsafe products, discrimination and environmental pollution.

5.3 Business Plan

A business plan is a written document describing a business idea, its customers, products or services, operations, marketing, resources, costs, expected income and goals. It helps an entrepreneur decide whether an idea is practical and communicate the plan to others.

A business plan should be realistic. Sales forecasts are estimates, not guaranteed income. A good plan identifies risks and explains how the business may respond to them.

Key points

- Executive summary: a short overview of the business idea and purpose.
- Business description: name, ownership, location, product or service and objectives.
- Market analysis: target customers, customer needs, competitors and demand.
- Marketing plan: product, price, place and promotion.
- Operations plan: materials, suppliers, equipment, location, work process and quality control.
- Management plan: roles, responsibilities and skills needed.
- Financial plan: start-up costs, operating costs, pricing, sales forecast, cash flow and expected surplus or loss.
- Risk analysis: possible problems such as low demand, price changes, theft, delays or equipment failure, with responses.

Business plan section	Questions to answer
Business idea	What will be sold or provided and what need will it meet?
Customers and market	Who will buy it and why? Who else supplies it?
Marketing	How will customers learn about it? How will price and delivery be decided?
Operations	Where and how will the product be made or service delivered?
Resources	What people, materials, tools and equipment are required?
Finance	What are the costs, selling price, expected sales and cash needs?
Risks	What could go wrong and how can the business reduce the risk?

Worked example / application

A learner plans to sell handmade bookmarks at a school event. Materials cost KSh 600. If 30 bookmarks are made and each sells for KSh 40, expected revenue is 30 × 40 = KSh 1,200. The simple surplus before any other costs is KSh 1,200 − KSh 600 = KSh 600. The plan should also consider unsold stock, labour, packaging, permission and time.

Practical activity

Write a one- to two-page business plan for a simple, safe school enterprise approved by the teacher. Include the business idea, target customers, competitor check, marketing approach, resources, start-up budget, selling price, sales estimate and risks.

Check your understanding

1. What is a business plan?

2. List seven sections of a business plan.

3. A business sells 25 items at KSh 80 each. Calculate expected revenue.

4. If the revenue is KSh 2,000 and recorded costs are KSh 1,350, calculate the simple surplus.

5. Why should a business plan include risks?

Suggested answers

1. A written description of a business idea, market, operations, resources, costs, income estimates and goals.

2. Executive summary, business description, market analysis, marketing, operations, management, financial plan and risk analysis.

3. 25 × 80 = KSh 2,000.

4. KSh 2,000 − KSh 1,350 = KSh 650.

5. To prepare for challenges and plan actions that reduce possible losses or disruption.

END-OF-YEAR REVISION

1. Explain four reasons why safety is important when working on raised platforms.

2. Describe six precautions when handling hazardous substances.

3. Explain how self-exploration supports career development.

4. Draw and label a simple rectangular block in oblique projection. Identify width, height and depth.

5. Differentiate sequence, selection and iteration in visual programming.

6. Write an algorithm for a program that asks for a learner's score and displays whether it is at least 50.

7. State five properties considered when selecting timber.

8. Explain the purpose of seasoning wood and identify four timber defects.

9. Describe the waste hierarchy and give one example at each of four stages.

10. Differentiate holding tools and driving tools, giving three examples of each.

11. Outline the stages of a design-and-make project.

12. Explain four services provided by financial institutions.

13. State five ways government influences business.

14. List the main sections of a business plan.

15. A business sells 45 products at KSh 120 each. Calculate revenue.

16. A business earns KSh 7,800 and has recorded costs of KSh 5,250. Calculate the simple surplus.

17. Explain why prototype testing and evaluation are important.

18. Give six pieces of evidence that should be included in a project portfolio.

Suggested answers

1. Prevent falls, injuries from falling objects, platform collapse and unsafe access; protect people and equipment.

2. Read labels, identify hazards, use ventilation and PPE, keep original containers, avoid mixing, secure storage and report spills.

3. It helps identify interests, strengths, values and areas for improvement so that career pathways can be researched and chosen thoughtfully.

4. Front face is true shape and size; parallel receding lines show depth at an angle, commonly 45 degrees. Label width, height and depth.

5. Sequence runs instructions in order; selection chooses between alternatives; iteration repeats instructions.

6. Start; input score; IF score ≥ 50 display 'At least 50' ELSE display 'Below 50'; end.

7. Grain, density, strength, durability, workability, moisture content and appearance.

8. Seasoning reduces moisture to reduce later shrinkage and warping; defects include knots, shakes, splits, warping, decay and insect attack.

9. Prevent/reduce waste (avoid unnecessary printing); reuse (use clean offcuts); recycle (send accepted paper/metal for recycling); recover or compost suitable material; dispose only as last resort.

10. Holding tools grip workpieces (vice, clamp, pliers); driving tools apply force to fasteners/materials (hammer, screwdriver, spanner).

11. Identify need, research, generate ideas, select design, plan, make, test, evaluate and communicate.

12. Savings, payments/transfers, loans/credit and insurance.

13. Infrastructure, laws, standards, taxation, licensing, training, consumer protection and enterprise support.

14. Executive summary, business description, market analysis, marketing plan, operations, management, financial plan and risk analysis.

15. 45 × 120 = KSh 5,400.

16. KSh 7,800 − KSh 5,250 = KSh 2,550.

17. Testing checks whether the product meets requirements; evaluation identifies strengths, weaknesses and improvements.

18. Problem statement, research, sketches, working drawing, materials/tool list, safety plan, cost estimate, progress record, test results and evaluation.

Glossary

Term	Meaning
Algorithm	A finite ordered set of steps for solving a problem.
Business plan	A written plan describing a business idea and how it will operate.
Career pathway	A route of education, training and work leading to career opportunities.
Debugging	Finding and correcting errors in a program.

Hazardous substance	A material that can harm people, property, animals or the environment.
Iteration	Repeating a set of instructions in a program.
Oblique projection	A pictorial drawing with the front face in true shape and angled receding depth lines.
Prototype	An early model used to test and improve a design.
Raised platform	A working surface above the surrounding ground or floor.
Selection	A programming structure that chooses an action based on a condition.
Seasoning	Controlled drying of timber to reduce moisture content.
Workpiece	The material being worked on.

Curriculum reference and review

Primary reference: Kenya Institute of Curriculum Development (KICD), Pre-Technical Studies Grade 9 Curriculum Design. Official PDF: https://kicd.ac.ke/wp-content/uploads/2024/06/Pre-Technical-Studies-Grade-9.pdf

The KICD Grade 9 design summary lists 120 suggested lessons across the 12 sub-strands included in these notes. Schools should check the current version and any rationalised updates used locally. The resource is teacher-prepared and should be reviewed by a Pre-Technical Studies teacher before classroom publication, especially practical safety procedures, visual programming tools and locally applicable business requirements.

End of Grade 9 Pre-Technical Studies Detailed Learner Notes.